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Field-Wide Evaluation of HCl and Viscoelastic Diverting Acid Systems for Carbonate Matrix Acidizing: Integrated Laboratory, Simulation, and Multi-Well Case Study

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Abstract

This study presents a comprehensive, field-scale evaluation of matrix acidizing treatments performed on four carbonate gas wells in southern Iran, integrating laboratory optimization, predictive simulation, and standardized field execution. The acid systems—comprising 28 wt% hydrochloric acid (HCl), 15 wt% viscoelastic diverting acid (VDA), and tailored additive packages—were systematically designed through dissolution capacity, emulsion compatibility, corrosion resistance, iron control, and H₂S scavenging tests under reservoir-simulated conditions. Numerical simulations guided the design of injection rates, stage volumes, and diversion sequences, ensuring acid placement efficiency while maintaining operations below fracture pressure. Field application delivered substantial performance gains, including gas production increases of up to 100%, wellhead pressure improvements of up to 84%, and skin factor reductions from as high as 5.0 to as low as −2.3. The thermo-reversible gelation behavior of VDA enabled effective diversion into low-permeability zones, promoting uniform wormhole distribution and enhanced reservoir contact. Post-treatment, Basic Sediment and Water (BS&W) content decreased significantly, further improving produced-fluid quality. Unlike conventional single-well analyses, this study offers a rare multi-well perspective, demonstrating the scalability, repeatability, and robustness of a unified HCl–VDA workflow in high-temperature, sour, and heterogeneous carbonate formations. The findings provide a transferable framework for optimizing acid placement and achieving predictable stimulation outcomes in similar carbonate reservoir settings worldwide.

Keywords: Matrix Acidizing, Viscoelastic Diverting Acid (VDA), Acidizing Operation, Simulation, Formation Damage, Gas Recovery

Introduction

Reservoir stimulation is a fundamental procedure in the oil and gas sector, primarily intended to improve the productivity of wells (1–3). Among the various stimulation techniques, matrix acidizing is a widely used

method, wherein acid is injected into the formation at pressures below the fracturing threshold (4, 5). This approach facilitates the removal of near-wellbore formation damage and the creation of new flow pathways, thereby enhancing permeability and improving hydrocarbon flow efficiency (6–8). By mitigating pressure losses and increasing output rates, matrix acidizing plays a pivotal role in restoring and maintaining well performance (9). In carbonate reservoirs, hydrochloric acid (HCl) is commonly utilized due to its strong reactivity with carbonate minerals such as calcium carbonate (CaCO_3) (10). Upon injection, HCl dissolves the rock matrix, promoting the formation of conductive channels that significantly improve fluid movement within the reservoir (11). To optimize the efficiency of acidizing treatments, specialized additives are often introduced (12, 13). Viscoelastic diverting acid (VDA) is a key formulation that incorporates additives designed to adjust the fluid's rheological behavior (14, 15). VDA is particularly effective because of its shear-sensitive viscosity: it exhibits high viscosity when static, aiding in flow diversion, and lowers its viscosity under shear, enabling smooth transport through the wellbore and into the formation (16). This viscoelastic behavior is crucial for ensuring effective acid distribution, particularly in heterogeneous formations with permeability variations (17). VDA can temporarily block high-permeability zones by forming viscous bridges, thereby directing the acid into less accessible, under-treated areas of the reservoir (18). Such targeted diversion results in more uniform stimulation and minimizes the risk of over-acidizing previously treated sections (19). The design and selection of acid additives in VDA systems must account for specific formation conditions and fluid characteristics. However, combining multiple additives requires careful compatibility assessments, as certain mixtures may react adversely with reservoir materials and cause unintended formation damage. Because reservoir properties vary widely—especially in terms of mineralogy and fluid composition—no universal additive formulation exists. Therefore, it is essential to test a range of commercial additives, including corrosion inhibitors, surfactants, solvents, and other functional agents, to tailor the acid system to the reservoir's unique requirements. Proper selection and use of these additives enhance the effectiveness and durability of acidizing treatments, ensuring improved well performance and long-term operational success (20, 21).

Matrix acidizing remains a cornerstone technique in the oil and gas industry, aimed at improving hydrocarbon recovery by minimizing formation damage and enhancing fluid conductivity within the reservoir matrix. Researchers have extensively explored innovative fluid formulations and diversion mechanisms to advance this stimulation method. Letichevsky et al. (2017) presented an innovative technology aimed at enhancing oil recovery in depleted carbonate reservoirs in Samara, Russia. They employed a foam acid system stabilized with a non-damaging agent to generate 65% nitrogen foam at the wellbore, which significantly improved and restored the natural flow of the wells (22). Furthermore, Gao et al. (2019) introduced the concept of "Large Liquid Volume Deep Penetration Acidizing" for carbonate reservoirs. This technique focused on generating long, stable wormholes that extended deeper into the formation, thereby increasing stimulation effectiveness (23). Al-Rekabi et al. (2020) further advanced the field by conducting a comprehensive study on carbonate matrix acidizing, with particular focus on skin evolution and fluid system optimization. Their methodology integrated real-time monitoring of skin behavior with optimized fluid compositions to reduce induced damage. The study resulted in measurable performance gains, including skin improvements between +3 and +10, and overall stimulation efficiencies in the range of 30% to 40%. Their work underscores the necessity of addressing critical factors such as soaking duration and acid system compatibility in achieving reliable stimulation outcomes (24). Elsheikh et al. (2022) evaluated the effectiveness of organic acids for stimulating the Sarah sandstone formation in Saudi Arabia, with the objective of identifying alternatives to the commonly used HCl-HF system in matrix acidizing of sandstone reservoirs. They conducted core flooding tests using organic acid blends of $\text{C}_2\text{H}_4\text{O}_2$ -HF and $\text{C}_2\text{H}_2\text{O}_4$ -HF on Berea sandstone cores to determine the optimum acid volume (12). Alvarez et al. (2022) reported on a novel foamed acid system designed for matrix stimulation of low-pressure, highly fractured carbonate formations in northern Iraq. They utilized stable foaming acids with foam qualities ranging from 60% to 80%, enabling successful treatment operations while eliminating the

risk of acid separation in aqueous environments (25). Shajari and Malhani (2023) discussed the challenges of conventional matrix acidizing operations in certain low-pressure oil fields in Iran. They investigated the use of a three-phase emulsified acid system—comprising hydrochloric acid, diesel, and nitrogen—for stimulation of low-pressure carbonate reservoirs in southern Iran (26). Keihanikamal et al. (2024) applied a three-stage approach combining laboratory analysis, simulation, and field implementation to optimize HCl and VDA treatments in carbonate and dolomite reservoirs of a southern Iranian gas field. Their work yielded a 100 % increase in gas production, an 84 % rise in gas pressure, and a skin factor reduction from 5 to -1.89 , underlining VDA's effectiveness in targeting low-permeability layers through thermo-reversible gelation (27). Mahdavi Kalatehno et al. (2025) carried out a detailed case study on carbonate matrix acidizing in a southern Iranian oil field, integrating laboratory tests, Design of Experiments/Response Surface Methodology simulations, and field operations. Their optimization of hydrochloric acid (HCl) and viscoelastic diverting acid (VDA) systems for high-temperature, low-permeability formations resulted in an 80 % increase in oil production, a 73 % rise in well pressure, and a reduction of the skin factor to 0.399 (28). This study presents a field-scale, multi-scenario evaluation of matrix acidizing operations using a unified HCl-VDA formulation, which marks a significant departure from conventional single-well investigations. The innovation lies in the consistent application of a chemically optimized acid system across four wells under varying operational conditions, allowing for a comparative assessment of treatment efficiency and fluid behavior. Unlike prior studies that focus narrowly on a single field case, this work captures the broader dynamics of acid distribution, diversion performance, and production response across different wells. The integration of lab-based formulation, predictive simulation, and real-time operational feedback enables a holistic and scalable framework for carbonate reservoir stimulation. These findings offer a practical and transferable methodology for engineers seeking to implement acidizing strategies in similar field settings with improved predictability and operational control.

Methods

Laboratory Evaluation

A comprehensive laboratory program was conducted to optimize acidizing treatments through systematic evaluation of acid–rock interactions, fluid compatibility, and additive performance under simulated reservoir conditions. Dissolution tests on composite carbonate samples (limestone and dolomite) assessed hydrochloric acid (HCl) at 7.5%, 15%, and 28% concentrations at 212 °F for six hours, with and without viscoelastic diverting acid (VDA), to determine optimal acid strength and the influence of additives including corrosion inhibitors, intensifiers, anti-emulsifiers, iron control agents, surfactants, and H₂S scavengers (7, 29). Emulsion compatibility was tested by blending acid formulations with solids and condensate under high-shear and high-temperature conditions to identify the minimum anti-emulsifier dosage required for complete phase separation within 30 minutes, including tests with spent acid (30). Iron control efficiency was evaluated using synthetic ferric solutions (500–1000 ppm Fe³⁺) aged at 212 °F to determine additive concentrations preventing ferric hydroxide precipitation across pH 2–4 (7, 31). Corrosion resistance was measured via steel coupon weight loss at 212 °F in acid solutions containing inhibitors and intensifiers, comparing main acid and VDA fluids for metal degradation (Table 1) (32–35). Hydrogen sulfide neutralization capacity was assessed using a modified Xiao et al. setup, wherein HCl reacted with FeS to generate H₂S, which was then contacted with scavenger solutions (0.5–1 wt%) and residual gas trapped in cadmium sulfate for gravimetric quantification of CdS precipitate, determining scavenger efficiency across varying temperatures (ambient and 90 °C) and pH (unbuffered and pH 4.5) (36, 37). Results collectively informed optimal acid formulation, additive selection, and operational parameters for field implementation.

Table 1—Dimensions and weight of corrosion coupons before corrosion test

	Hole Diameter (cm)	Length (cm)	Width (mm)	Thickness (mm)	Coupon Initial Weight (gr)
Main Acid	0.598	7.585	2.194	0.161	21.520
VDA	0.593	7.594	2.2	0.166	21.470

Simulation

Prior to field implementation, a comprehensive simulation workflow was developed to optimize the design and operational parameters of the acidizing treatments. The model was used to refine injection sequences, flow rates, and pressure limits, and to improve understanding of the governing mechanisms of acid stimulation. This framework integrated well architecture, reservoir properties, skin factor measurements, surface injection constraints, and chemical treatment protocols, including diversion and pumping strategies. Key well parameters used in the modeling process are summarized in [Table 2](#).

Table 2—Well data for simulation

Segment Type	Well 1	Well 2	Well 3	Well 4
Top of Open Section (ft TVD)	8771	8668	8636	8708
Bottom of Open Section (ft TVD)	9483	9524	9607	9616
Initial Skin	1.95	5	5	3
Max. Surface Pressure (psi)	1750	1620	1614	1828
Fracture Pressure (psi)	6388	6364	6391	6096

The simulation software was employed to model acid–carbonate interactions, with emphasis on wormhole propagation and pressure evolution, enabling real-time monitoring of flow behavior and operational parameters for dynamic adjustment of the pumping schedule. This capability was critical for maintaining simulated pressures below the fracture pressure limit, as defined by field-measured fracture gradients. A treatment schedule, informed by laboratory results, was developed to define chemical formulations and phase sequences—including preflush, main acid, viscoelastic diverting acid (VDA), overflush, and final displacement—with precise control of additive volumes and concentrations. The resulting model accurately replicated downhole conditions, provided a reliable basis for operational planning, and ensured that all acidizing operations were executed within safe pressure margins to preserve reservoir matrix integrity and prevent formation damage.

Operation Procedure

This project focused on the matrix acidizing of a gas well located in a gas field in southern Iran. Four wells with comparable reservoir properties were evaluated, all of which were completed in carbonate formations predominantly composed of dolomite. The primary objective of the acidizing treatment was to enhance gas production by improving reservoir permeability and eliminating formation damage or flow restrictions. The operation was executed using a stimulation vessel to prepare and inject a specially designed acid system into the reservoir. This system consisted of multiple stages, including pre-flush, main acid, diverter, and overflush, each selected based on laboratory experiments and numerical simulations. The detailed well and reservoir characteristics, along with the acidizing design parameters, are summarized in [Table 3](#).

Table 3—Well Data

	Well No. 1	Well No. 2	Well No. 3	Well No. 4
Well Type	Inclined	Vertical	Vertical	Vertical
Formation	Layers A, B, C	Layers A, B, C, D	Layers A, B, C, D	Layers A, B, C, D
Reservoir Rock Type	Dolomite & Limestone	Dolomite & Limestone	Dolomite & Limestone	Dolomite & Limestone
H ₂ S (ppm)	1800	6000	6000	5000
CO ₂ (%)	1	1.8	2	1.5
Total Interval (ft)	518.4	679.2	907.2	797.3
Reservoir Pressure (psi)	A: 4300 B: 3200 Top of C: 3200 Bottom of C: 4000	A: 3450 B: 3000 C: 3000 D: 3150	A: 3150 B: 3000 C: 3100 D: 3000	A: 3200 B: 2905 C: 3400 D: 3155
Reservoir Temperature (F)	210	220	220	210
Fracture Gradient (psi/ft)	0.7	0.7	0.7	0.7

Results and discussion

Laboratory Tests

In the laboratory phase of this study, a main acid formulation and a viscoelastic diverting acid (VDA) system were specifically designed and optimized under reservoir-simulated conditions. These acid systems, enhanced with specialized additives, were developed to be applicable and were subsequently used for all four operational wells in the field phase.

Main Acid.

Dissolution Test

The main purpose of the dissolution test was to see how well an acid formulation designed for acidizing operations worked, with a focus on carbonate and dolomite reservoir materials. The acid formulation consisted of hydrochloric acid and various additives, as shown in Table 4. The dissolution test was conducted to measure the amount of limestone dissolved by the acid solution at various concentrations and temperatures.

Table 4—List of ingredients in acid solution

Contract Description	Quantity (ml)
Hydrochloric Acid	450
Corrosion Inhibitor	8
Corrosion Inhibitor Intensifier	5
Non-Emulsifier	3
Iron Contorol	25
Surfactant	4.5
H ₂ S Scavenger	4.5
Total	500

The 28 wt% HCl solution achieved nearly complete dissolution of limestone at both tested temperatures, confirming its superior applicability for carbonate and dolomite acidizing. Lower concentrations exhibited reduced dissolution, particularly at lower temperatures, indicating limited wormhole penetration potential.

The high performance of 28 wt% HCl is attributed to its strong reactivity with CaCO_3 , generating CO_2 to enhance acid penetration. Additives—including corrosion inhibitors, non-emulsifiers, iron-control agents, surfactants, and H_2S scavengers—further optimized performance by mitigating corrosion, preventing emulsion and iron precipitation, improving wettability, and controlling sour gas. The tested formulation thus represents a robust candidate for high-efficiency stimulation under varied temperature conditions.

Emulsion Test

Emulsion tests across condensate-to-acid ratios of 25:75, 50:50, and 75:25, for both live and spent acid, confirmed the anti-emulsion additive's effectiveness, achieving complete phase separation in under three minutes with no emulsion formation. The optimal concentration was determined as 3 ml, below which separation times exceeded 30 minutes. This low dosage requirement reduces chemical consumption, operational cost, and environmental footprint, while maintaining consistent performance across varying fluid ratios and acid conditions.

Iron Control Test

The Iron Control Test was conducted to assess the effectiveness of iron control additives in mitigating iron precipitation. In the pursuit of optimizing the performance of these additives, the study delved into the impact of varying the concentrations of the iron control. This involved a systematic exploration, ranging from 0 to 25 ml. The results unequivocally pointed to the efficacy of 25 ml of iron control as the optimal concentration. This concentration was found to effectively sustain the solubility of iron ions throughout the duration of the test. This outcome is pivotal, as it establishes a clear and efficient dosage guideline for the use of iron control additives to prevent undesirable iron precipitation (Figure 1).

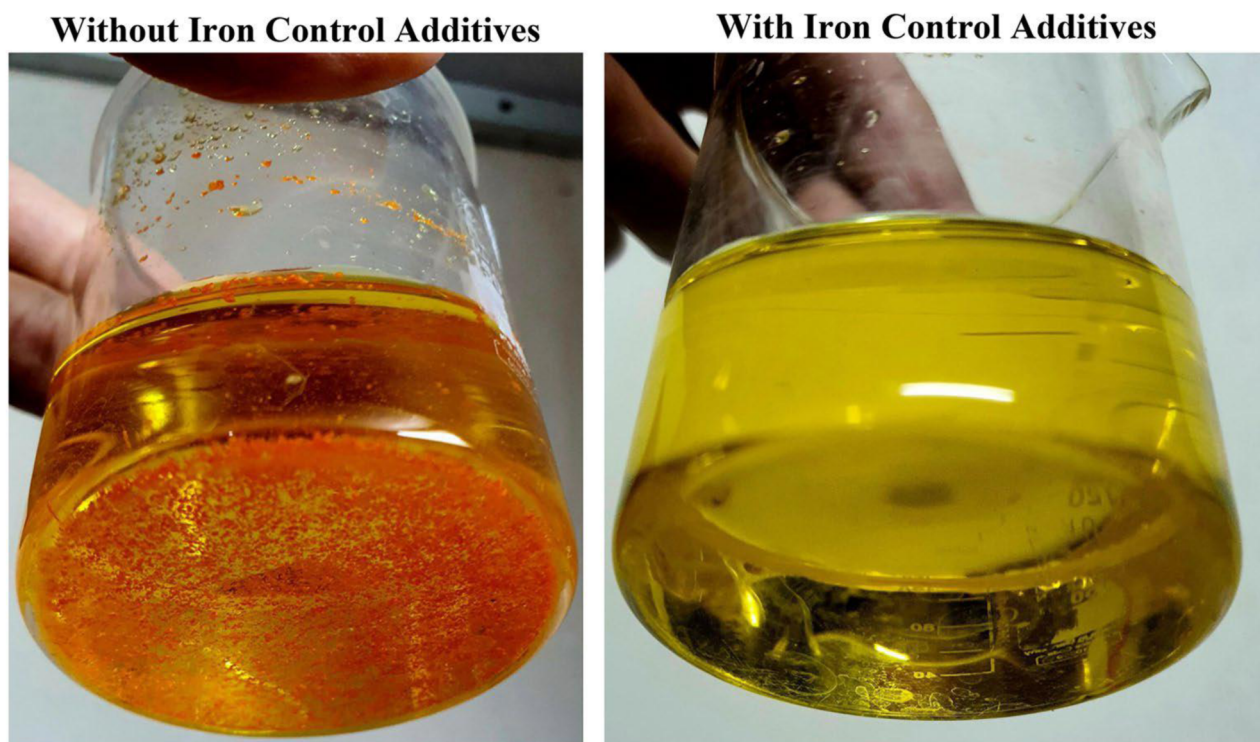


Figure 1—Iron Control Test (27)

Corrosion Test

In this section, a 6-hour corrosion test was carried out at a temperature of 212 degrees Fahrenheit to assess the potential impact of corrosion inhibitors and corrosion inhibitor intensifiers on the corrosion rate of the main acid. The primary objective was to determine if these chemical additives could effectively reduce the corrosion rate, with the stringent criterion of keeping it below 45 millimeters per year over a six-day period.

The results of this investigation, including weight loss, time, surface area, corrosion sample density, and corrosion rate, are presented in [Table 5](#).

Table 5—Corrosion results for the main acid

Coupon Weight after test (gr)	W_{loss} (gr)	ρ (gr/cm ³)	A (cm ²)	T (hr)	CR (mm/y)
21.223	0.297	8.31	16.36	6	31.9

Following the addition of 8 ml of corrosion inhibitor and 5 ml of the Corrosion Inhibitor Intensifier, the corrosion rate decreased to approximately 31.9 millimeters per year. Although this value falls in close proximity to the permissible limit of 45 millimeters per year, it still falls within an acceptable range. This outcome suggests that the application of 8 ml of corrosion inhibitor and 5 ml of the Corrosion Inhibitor Intensifier may potentially serve as an effective strategy for corrosion mitigation in acidizing operations. It is noteworthy that acid inhibitors are typically highly concentrated compounds. Therefore, the reduction in the quantity of corrosion inhibitors, as demonstrated in this study, may offer significant cost savings for acidizing operations. Consequently, the findings not only underscore the potential effectiveness of the recommended dosages but also highlight their potential cost-effectiveness as a viable strategy for corrosion reduction in acidizing operations.

H₂S Scavenging Test

H₂S scavenger efficiency was evaluated via CdS precipitate quantification from FeS–HCl reactions. Increasing scavenger concentration reduced CdS formation, with an optimal dose of 4.5 ml yielding a minimal precipitate mass of 0.02 g and a calculated scavenging efficiency of 92%. These results confirm the scavenger's critical role in mitigating H₂S release during matrix acidizing, enhancing safety and operational reliability ([Figure 2](#)).



Figure 2—Precipitate of CdS @ Trapping Cylinders

Viscoelastic Diverting Acid (VDA).

Dissolution Test

The VDA Dissolution Test is a valuable method for evaluating the performance of Viscoelastic Diverting Acid (VDA) in dissolving carbonate and dolomite reservoir materials. VDA, a polymer-free acidizing fluid, is designed to boost carbonate stimulation. Solubility experiments were conducted using VDA to assess its effectiveness. This section presents the results of the VDA Dissolution Test and provides an analysis of their implications for carbonate reservoir stimulation. The test was performed to examine the influence of VDA on carbonate and dolomite reservoir materials. A specialized VDA formulation was prepared, containing a

15% concentration of hydrochloric acid. Furthermore, this formulation included specific additives intended to enhance VDA's performance, as outlined in Table 6.

Table 6—List of ingredients in VDA solution

Contract Description	Quantity (ml)
Hydrochloric Acid	440
Corrosion Inhibitor	7
Corrosion Inhibitor Intensifier	5
Non-Emulsifier	5
Iron Control	6
Methanol	7
Visco- Elastic Surfactant	5
Total	475

VDA dissolution tests at surface temperature and 212°F, over two hours, revealed that live VDA remained fully soluble without initiating mineral reaction, indicating stable pre-reaction fluidity upon injection. In contrast, spent VDA at 212°F transitioned to a gel-like state, increasing viscosity and enabling temporary blockage of high-permeability channels. This diversion effect promotes deeper acid penetration, improving stimulation efficiency in heterogeneous carbonate and dolomite formations. These findings affirm VDA's value as an effective in-situ diverting agent for enhanced reservoir contact.

Figure 3 illustrates VDA's thermo-reversible gelation after rock contact. At room temperature, viscosity rapidly increased to ~400 cP within 10 minutes, forming a temporary gel that restricts flow in high-permeability zones and redirects fluids toward lower-permeability intervals. With increasing temperature, gel structures degraded, reducing viscosity to a minimum near 212 °F—close to reservoir conditions—allowing the fluid to regain mobility and flow back without impairing productivity. This reversible behavior enables temporary diversion with minimal risk of formation damage in high-temperature reservoirs.

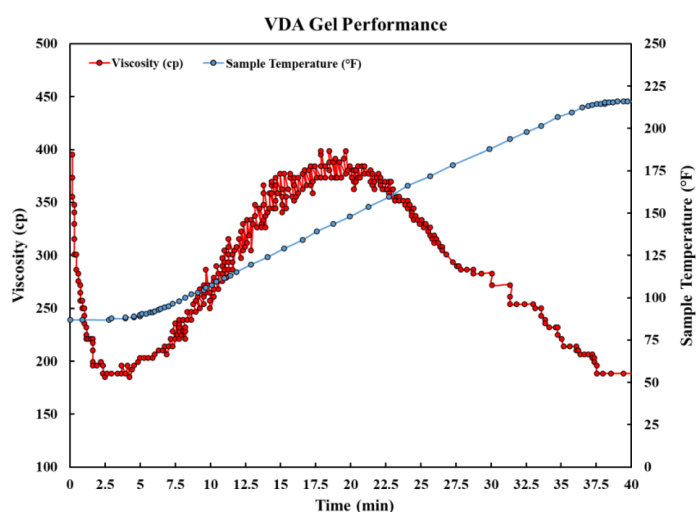


Figure 3—VDA Gel Performance

Emulsion Test

Emulsion tests with condensate-to-VDA ratios of 25:75, 50:50, and 75:25, for both live and spent VDA, showed complete phase separation within 3 minutes and no emulsion formation at an optimal additive

dosage of 5 ml. Lower dosages prolonged separation beyond the 30-minute benchmark. The additive demonstrated consistent effectiveness across all tested ratios and acid conditions, ensuring rapid separation and stable fluid handling during operations.

Corrosion Test

In this section, the focus shifts to an examination of the corrosion rate of VDA in the presence of corrosion inhibitors and Corrosion Inhibitor intensifiers. The corrosion test was rigorously executed at a temperature of 212 degrees Fahrenheit for a duration of six hours, adhering to the manufacturer's stipulation of a maximum allowable corrosion rate of 45 millimeters per year.

Table 7 provides a thorough display of the variables used to calculate the precise corrosion rate. Of particular interest, the inclusion of 7 ml of corrosion inhibitor and 5 ml of Corrosion Inhibitor Intensifier resulted in a notably low corrosion rate of approximately 5.89 millimeters per year, well within the manufacturer's established limits. Furthermore, the corrosion results reveal a compelling observation: VDA demonstrates a considerably diminished corrosion rate when contrasted with the primary acid. This shows that corrosion inhibitors and intensifiers are very good at keeping corrosion well below acceptable levels. It also shows that VDA is better at stopping corrosion than the primary acid.

Table 7—Corrosion results for the VDA

Coupon Weight after test (gr)	W_{loss} (gr)	ρ (gr/cm ³)	A (cm ³)	T (hr)	CR (mm/y)
21.417	0.053	8	16.43	6	5.89

Simulation and Operation

Acidizing Composition Design. This section outlines the design of the acidizing treatment sequence, including preflush, main acid, diverting agent, and postflush stages. It is important to note that the total volume of each stage is determined based on the thickness of the targeted reservoir zone and is calculated using the software, and established correlations as well as laboratory test results. Table 8 presents the volume parameters for each stage of the treatment across different wells.

Table 8—Acidizing volume for all wells

Stage Type	Injection Volume (gal)			
	Well 1	Well 2	Well 3	Well 4
Pre-flush	3629	4754	6350	5581
Main Acid (28 %)	36288	47542	63504	47837
VDA (15 %)	36288	36977	63504	47837
VDA (20 %)	0	10565	0	0
Over-flush	6221	8150	10886	9568
Displacement	17640	16872	21162	16949

The preflush stage removed brine and conditioned the formation using a solvent-based formulation (methanol, isopropyl alcohol), surfactant, and H₂S scavenger. The solvents dissolved condensate films and mitigated water blockage, while the surfactant reduced interfacial tension and altered wettability toward water-wet conditions, improving displacement efficiency. The H₂S scavenger reduced sour gas content and prevented FeS precipitation. Additive ratios were optimized through laboratory tests on reservoir-specific rock and fluid samples (Table 9).

Table 9—Preflush composition for all wells

Stage Type	Injection Volume (gal)			
	Well 1	Well 2	Well 3	Well 4
Mixing Water	2649	3471	4636	4074
Surfactant	36	48	64	56
ISO Propanol	181	238	318	279
Methanol	363	475	635	558
Mutual Solvent	363	475	635	558
H ₂ S Scavenger	36	48	64	56

In carbonate formations, the main acid stage is designed to stimulate the reservoir by creating wormholes, enabling penetration through the damaged zone around the wellbore and reducing the skin factor. Based on laboratory tests, 28% hydrochloric acid (HCl) was selected as the main acid. The acid solution also included several additives such as corrosion inhibitors, corrosion inhibitor intensifiers, demulsifiers, iron control and sequestrant agents, surfactants, and H₂S scavengers. Corrosion inhibitors were used to protect surface equipment and production tubing from acid corrosion. Demulsifiers were added to prevent emulsion formation, which could hinder the acid-rock reaction. Iron control agents were used to regulate the pH and prevent the precipitation of iron compounds, which can clog pore spaces. The specific acid composition used in each well is presented in Table 10.

Table 10—Main acid composition for all wells

Stage Type	Injection Volume (gal)			
	Well 1	Well 2	Well 3	Well 4
HCl Acid	31135	40791	54486	41044
Mixing Water	2903	3756	5017	3683
Corrosion inhibitor	544	713	953	718
Corrosion Inhibitor Intensifier	363	475	635	478
Non-Emulsifier	254	380	508	335
Iron Sequestering Agent	363	475	535	478
Iron Chelating Agent	1814	2377	3175	2392
Surfactant	363	475	635	478
H ₂ S Scavenger	363	475	635	478

The viscoelastic diverting acid (VDA) is a polymer-free acidic fluid that can be used either as a standalone stage or in conjunction with other acids to stimulate all areas of a carbonate reservoir. Upon contact with the formation, VDA forms a gel-like solution that increases viscosity under reservoir conditions. This allows it to temporarily block high-permeability zones, diverting the remaining acid to lower-permeability zones, thereby enhancing areal coverage in heterogeneous formations. The use of VDA promotes efficient wormholing under varying conditions, enhances fluid efficiency during acidizing operations, and improves reservoir contact. The VDA composition used in each well is listed in Table 11.

Table 11—Diverter composition (VDA 15%) for all wells

Stage Type	Injection Volume (gal)				
	Well 1	Well 2		Well 3	Well 4
		15%	20%		
HCl Acid	15713	16011	6244	27497	20713
Mixing Water	16148	16344	3000	28386	21287
Corrosion inhibitor	435	444	127	762	574
Corrosion Inhibitor Intensifier	290	296	85	508	383
Non-Emulsifier	254	185	53	318	335
Iron Sequestering Agent	363	370	106	635	478
Methanol	363	370	106	635	478
Visco-Elastic Surfactant	2722	2958	845	4763	3588

The VDA solution includes a viscoelastic surfactant, a cationic additive that increases the viscosity of hydrochloric acid and imparts a more linear flow profile to the acid mixture. This helps control acid placement and penetration into the formation, while also minimizing the risk of insoluble particle precipitation in pore spaces. The objective of the postflush stage is to displace the main acid away from the near-wellbore region and to push precipitates formed during acidizing operations further into the formation. The postflush solution, typically composed of water or brine, helps minimize acid contact time with production tubing and casing. Table 12 presents the postflush compositions used in different wells.

Table 12—Postflush composition for all wells

Stage Type	Injection Volume (gal)			
	Well 1	Well 2	Well 3	Well 4
Mixing Water	4914	6439	8600	7558
Surfactant	62	82	109	96
Methanol	622	815	1089	957
Mutual Solvent	622	815	1089	957

The use of VDA necessitated the inclusion of diesel and a common solvent in the postflush stage. This combination helps dissolve any emulsions or gel clusters that may form due to the relatively high surfactant concentration in the VDA formulation. Additionally, the postflush phase assists in dissolving residual corrosion inhibitors accumulated in the formation, which could otherwise reduce permeability.

Simulation. Stimpro 2019 software was employed to simulate carbonate reservoir acidizing and optimize injection rates and stage numbers. Optimization relied on formation breakdown pressure and required fluid volume, derived from reservoir porosity, permeability, and damaged zone radius. Acid was injected below fracture pressure to avoid fracturing, promoting wormhole formation to bypass damage and improve productivity. Skin factor reduction results guided the software in determining optimal stage count and injection rates, complemented by field experience. Final optimized parameters per well are detailed in Table 13 to Table 16.

Table 13—Optimum number of stages and injection rate for Well No. 1

Step	Step Type	Fluid Volume (gal)	Flow Rate (bpm)
1	Preflush	3829	10
2	Main acid	7257	15
3	Diversion	9072	25
4	Main acid	7257	35
5	Diversion	9072	40
6	Main acid	7257	40
7	Diversion	9072	40
8	Main acid	7257	40
9	Diversion	9072	40
10	Main acid	7257	40
11	Postflush	6221	45
12	Displacement (Sea Water)	17640	45

Table 14—Optimum number of stages and injection rate for Well No. 2

Step	Step Type	Fluid Volume (gal)	Flow Rate (bpm)
1	Preflush	4754	7
2	Main acid	9508	7
3	Diversion	5282	7
4	Main acid	4754	15
5	Diversion	5282	20
6	Main acid	4754	25
7	Diversion	5282	25
8	Main acid	4754	25
9	Diversion	5282	25
10	Main acid	4754	25
11	Diversion	5282	25
12	Main acid	4754	25
13	Diversion	5282	25
14	Main acid	4754	25
15	Diversion	5282	25
16	Main acid	2377	25
17	Diversion (20%)	3521	25
18	Main acid	2377	25
19	Diversion (20%)	3521	25
20	Main acid	2377	25
21	Diversion (20%)	3521	25
22	Main acid	2377	25
23	Postflush	8150	25
24	Displacement (Sea Water)	16872	25

Table 15—Optimum number of stages and injection rate for Well No. 3

Step	Step Type	Fluid Volume (gal)	Flow Rate (bpm)
1	Preflush	6350	10
2	Main acid	7056	15
3	Diversion	7938	25
4	Main acid	7056	35
5	Diversion	7938	35
6	Main acid	7056	35
7	Diversion	7938	35
8	Main acid	7056	35
9	Diversion	7938	35
10	Main acid	7056	35
11	Diversion	7938	35
12	Main acid	7056	35
13	Diversion	7938	35
14	Main acid	7056	35
15	Diversion	7938	35
16	Main acid	7056	35
17	Diversion	7938	35
18	Main acid	7056	35
19	Postflush	7938	35
20	Displacement (Sea Water)	21162	40

Table 16—Optimum number of stages and injection rate for Well No. 4

Step	Step Type	Fluid Volume (gal)	Flow Rate (bpm)
1	Preflush	5581	10
2	Main acid	5980	15
3	Diversion	6834	25
4	Main acid	5980	35
5	Diversion	6834	40
6	Main acid	5980	40
7	Diversion	6834	40
8	Main acid	5980	40
9	Diversion	6834	40
10	Main acid	5980	40
11	Diversion	6834	40
12	Main acid	5980	40
13	Diversion	6834	40
14	Main acid	5980	40
15	Diversion	6834	40
16	Main acid	5980	40

Step	Step Type	Fluid Volume (gal)	Flow Rate (bpm)
17	Postflush	9568	45
18	Displacement (Sea Water)	16949	45

Field treatments adhered to simulation-based designs, with Maximum Surface Injection Pressure (MSIP) limits calculated from reservoir depth, fracture gradient, and hydrostatic pressure to prevent unintended fracturing. Each well received approximately 70 gal/ft of treatment volume—sufficient to propagate wormholes beyond damaged zones. A five-stage pumping schedule maintained pressures below fracture limits, with rates adjusted to injectivity and operational requirements. In the final stage, rates were increased to the maximum allowable without exceeding MSIP. Figure 4 for Well 1 illustrates pumping rate and pressure profiles, showing pressure fluctuations due to VDA diversion into lower-permeability layers. This diversion improved stimulation uniformity and enhanced permeability across targeted intervals.

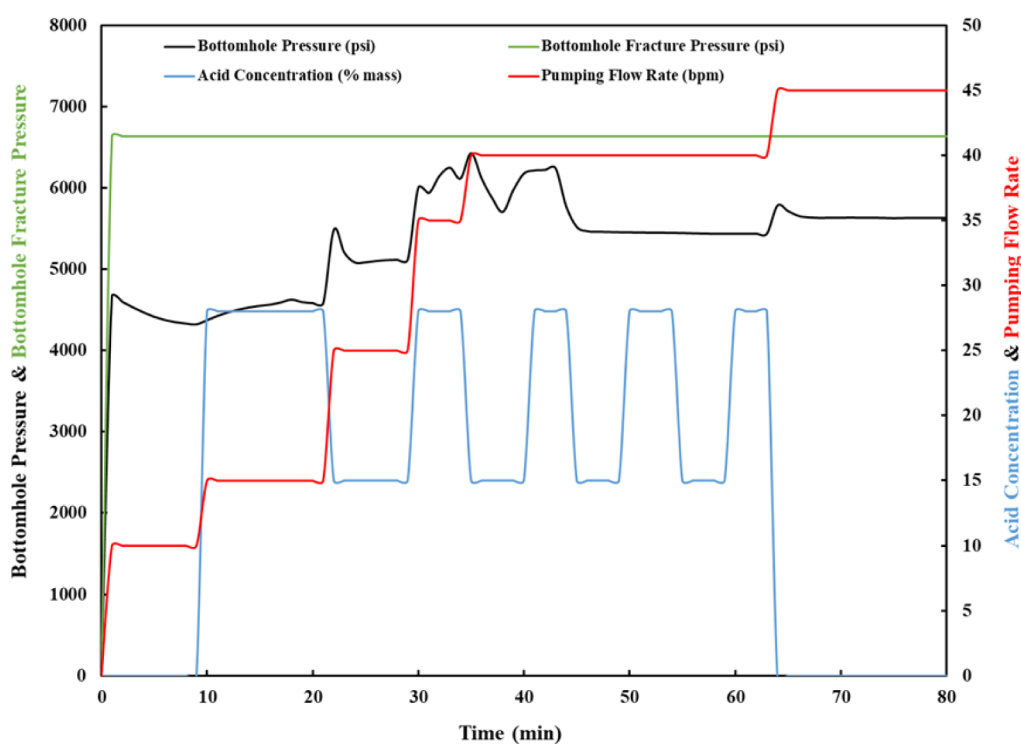


Figure 4—Pressure match during the acid injection (27)

Figure 5 shows progressive skin reduction during acidizing of well no. 1. Initial contact of the 28% HCl primary flush with formation minerals dissolves damage, lowering skin. Subsequent acid stages with viscoelastic diverting acid (VDA) enhance this effect by temporarily blocking high-permeability zones, diverting acid into tighter intervals, and improving stimulation efficiency. Similar patterns were observed for the other wells.

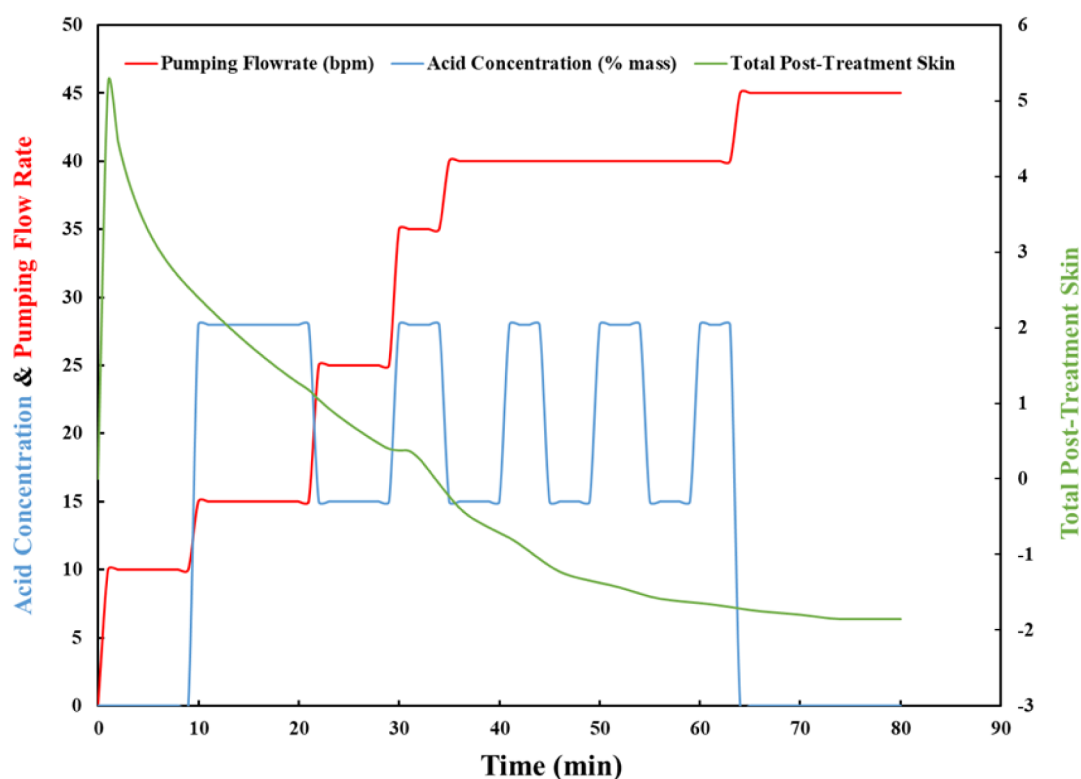


Figure 5—Skin reduction, pumping flow rate and acid concentration plot (27)

Furthermore, an examination of the acid invasion profile, as showcased in Figure 6, reveals a notable pattern. The majority of the viscoelastic diverting acid (VDA) follows a path into the layers characterized by higher permeability. After a brief period, this inflow leads to a surge in fluid viscosity. This specific occurrence facilitates the later treatment stages in reaching layers characterized by lower permeability. Because of this, all layers of the reservoir are stimulated, and the maximum radial penetration can reach up to 180, 185, 270 and 180 inches from the wellbore, for wells number 1 to 4 respectfully. The simulation results clearly show that the matrix acidizing treatment increases productivity by creating wormholes near the damaged area, which in turn increases the permeability of the formation. This treatment also exerts a pronounced influence on reducing the skin factor and enhancing flow efficiency. It's worth noting that the acidization of the matrix adheres to best practices and recommendations tailored to carbonate reservoirs.

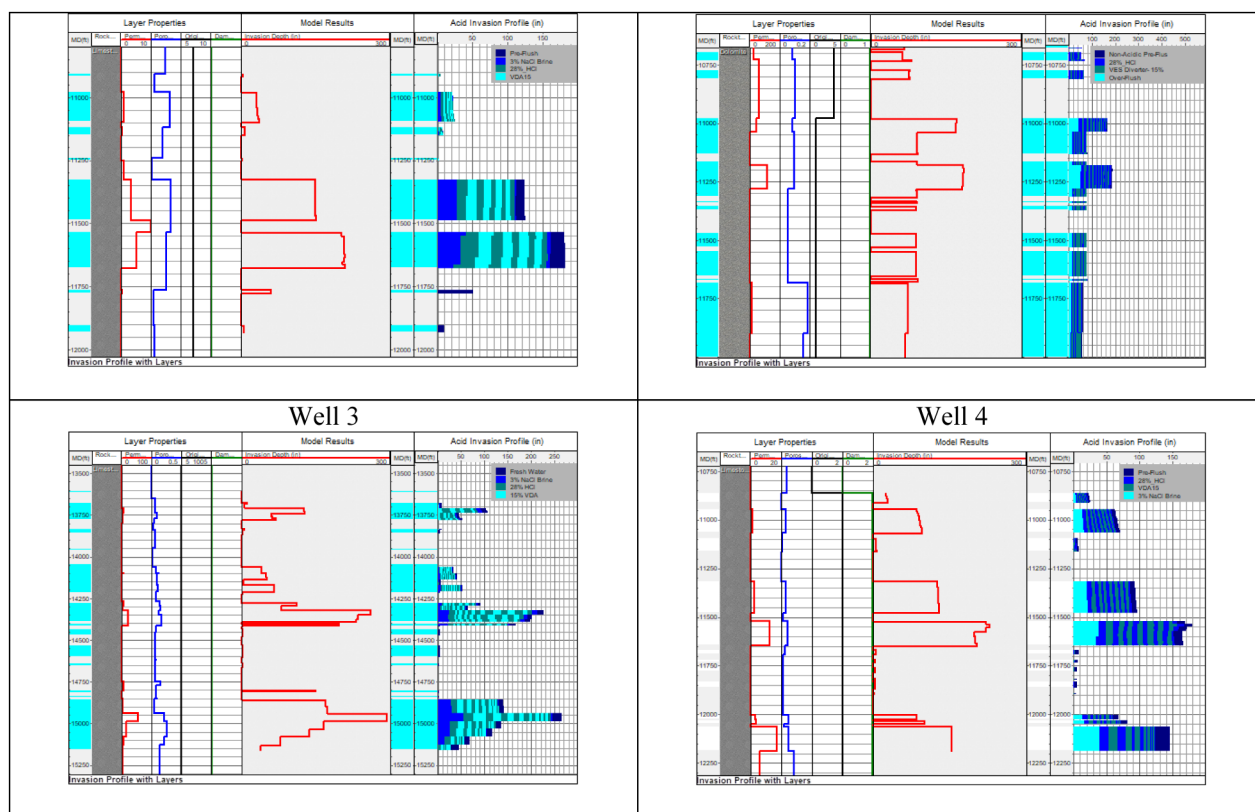


Figure 6—Acid invasion profile

Operation Results. Figure 7 presents the pressure and acid volume profile for well no. 1. The operation began at ~2,500 psi surface pressure, which dropped until acid reached the reservoir (~minute 22). VDA arrival caused pressure increases as high-permeability channels were temporarily blocked, forcing acid into lower-permeability paths. Stages 4–6 required rate reduction to avoid exceeding the 1,840 psi surface pressure limit and prevent fracturing. Real-time monitoring enabled adjustments, and after Stage 6, injection continued at the maximum safe rate. Post-treatment, well no. 1 was flowed back for 153 hours to record performance parameters.

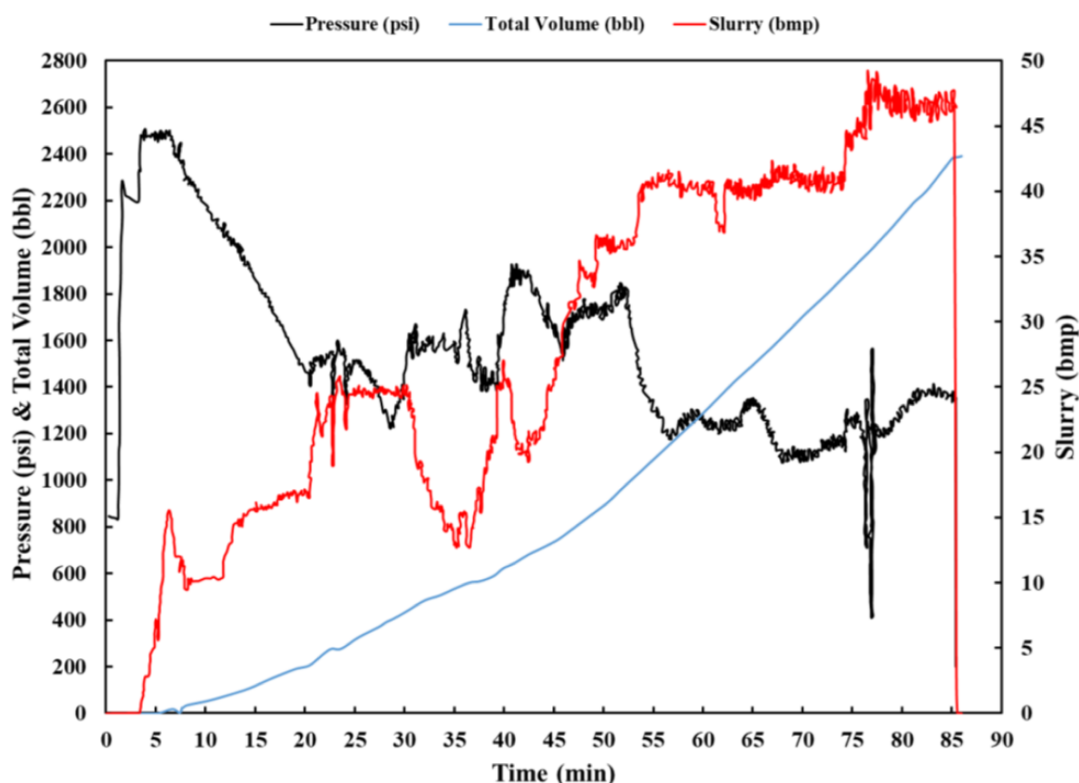


Figure 7—Pressure, Volume and Pump Rate changes during the acid injection for well no. 1 (27)

Table 17 shows that gas production from well no. 1 doubled after acidizing, with an 84% wellhead pressure increase (+907 psi) and a BS&W drop from 7% to 3%. These gains align with a skin factor reduction to -1.89 , confirming successful matrix stimulation and improved fluid quality. Comparable results were observed for other wells (Table 18 to Table 20).

Table 17—Production parameters of the well no. 1 (27)

Pre-Acid Clean Up	Post-Acid Clean Up
Choke Size: 72/64" fix	Choke Size: 72/64" fix
WHP: 1075 psia	WHP: 1982 psia
WHT: 132.8 °F	WHT: 149 °F
BS&W: 7%	BS&W: 3%
EsT. Rate: 26 MMSCF/D	EsT. Rate: 50 MMSCF/D NeT Flowing Time: 153 hrs
Skin: 5	Skin: -1.89

Table 18—Production parameters of the well no. 2

Pre-Acid Clean Up	Post-Acid Clean Up
Choke Size: 64/64" fix	Choke Size: 72/64" fix
WHP: 2451 psia	WHP: 2466 psia
WHT: 60 °C	WHT: 74 °C
BS&W: 14 %	BS&W: 12 %
EsT. Rate: 48.38 MMSCF/D	EsT. Rate: 61.44 MMSCF/D
Skin: 3	Skin: -1.1

Table 19—Production parameters of the well no. 3

Pre-Acid Clean Up	Post-Acid Clean Up
Choke Size: 64/64" fix	Choke Size: 72/64" fix
WHP: 2442 psia	WHP: 2458 psia
WHT: 73 °C	WHT: 76 °C
BS&W: 17 %	BS&W: 15 %
EsT. Rate: 47.53 MMSCF/D	EsT. Rate: 61.32 MMSCF/D
Skin: 4	Skin: -0.8

Table 20—Production parameters of the well no. 4

Pre-Acid Clean Up	Post-Acid Clean Up
Choke Size: 64/64" fix	Choke Size: 72/64" fix
WHP: 2579 psia	WHP: 2621 psia
WHT: 69 °C	WHT: 75 °C
BS&W: 16 %	BS&W: 12 %
EsT. Rate: 51.2 MMSCF/D	EsT. Rate: 65.2 MMSCF/D
Skin: 2.8	Skin: -2.3

All wells received the same acid compositions (28% HCl and 15% VDA), enabling direct comparison. The number of stages and injection rates varied due to geological and operational factors, with higher rates in later stages likely promoting deeper acid penetration. Across the field, post-acidizing results demonstrated increased wellhead pressures, reduced BS&W, and higher production rates, confirming the effectiveness of VDA-assisted carbonate matrix acidizing.

Conclusion

This study presents a comprehensive, multi-well evaluation of carbonate matrix acidizing using a unified hydrochloric acid (HCl) and viscoelastic diverting acid (VDA) workflow, integrating laboratory design, numerical simulation, and field-scale execution. The standardized acid system, optimized through dissolution, compatibility, corrosion, iron control, and H₂S scavenging tests, demonstrated excellent rock dissolution capacity, effective diversion, and minimal corrosion under high-temperature, sour-reservoir conditions.

Field implementation across four carbonate gas wells in southern Iran resulted in substantial productivity improvements, including up to 100% increases in gas production, wellhead pressure gains of up to 84%, and skin factor reductions to as low as -2.3. The deployment of VDA ensured efficient diversion into low-permeability zones via thermo-reversible gelation, leading to uniform stimulation and enhanced reservoir contact. Additional operational benefits included significant reductions in Basic Sediment and Water (BS&W) and improved produced-fluid quality.

The integrated approach—combining laboratory optimization, predictive simulation, and consistent operational protocols—proved scalable and repeatable, enabling reliable acid placement and predictable production response. These results validate the HCl-VDA system as a robust, field-ready solution for high-temperature, heterogeneous carbonate reservoirs, offering a transferable methodology for similar applications worldwide.

Future work should focus on long-term production monitoring, cost-benefit analysis, and potential integration with data-driven diagnostic tools to further enhance acidizing efficiency and decision-making in carbonate stimulation projects.

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References

1. Xiujian D, Guangdi L, Mingliang S, Pangen W. Origin of polygonal fault systems: A case from the Sanzhao sag in the Songliao Basin, East China. *Petroleum Exploration and Development*. 2013;**40**(3): 333–43.
2. Khamsehchi E DM, Imeri M, Mahdavi Kalatehno J, Khaleghi MR. Pipe Diameter Optimization and Two-Phase Flow Pressure Drop in Seabed Pipelines: A Genetic Algorithm Approach. *Interciencia* 2023. <https://doi.org/10.59671/9tuvf>.
3. Yousefmarzi F, Ahmadi M, Mahdavi Kalatehno J, Dargi M, Miladi K, editors. Integrating Artificial Neural Networks and Genetic Programming for Enhanced Gas Production Forecasting. *SPE EOR Conference at Oil and Gas West Asia*; 2025: SPE.
4. Zhang L, He J, Wang H, Li Z, Zhou F, Mou J. Experimental investigation on wormhole propagation during foamed-VES acidizing. *Journal of Petroleum Science and Engineering*. 2021;**198**: 108139.
5. Liu K, Zhong X, Zhu Y, Pan D, Li Z, Chen C. Experimental Study on the Influence of Acid-Pressure Compound Effect on Multi-Scale Pore Evolution of Oil Shale. *Arabian Journal for Science and Engineering*. 2022;**47**(6): 7419–32.
6. Dargi M, Khamsehchi E, Mahdavi Kalatehno J. Optimizing acidizing design and effectiveness assessment with machine learning for predicting post-acidizing permeability. *Scientific Reports*. 2023;**13**(1): 11851.
7. Kalatehno JM, Khamsehchi E, Abbasi A, Khaleghi MR. A novel approach to determining appropriate additive concentrations for stimulation of gas carbonate reservoirs. *Results in Engineering*. 2023: 101440.
8. Ridha S, Hamid AIA, Setiawan RA, Ibrahim MA, Shahari AR. Microstructure behavior of fly ash-based geopolymers cement exposed to acidic environment for oil well cementing. *Arabian Journal for Science and Engineering*. 2018;**43**: 6413–28.
9. Khalil R, Emadi H, Altawati F. Investigating the effect of matrix acidizing injection pressure on carbonate-rich Marcellus shale core samples: an experimental study. *Journal of Petroleum Exploration and Production*. 2021;**11**: 725–34.
10. Khaleghi MR, Khamsehchi E, Abbasi A, Kalatehno JMJCJoCE. Evaluation of the effectiveness of mixing hydrochloric acid and organic acid as stimulation fluids in a tight gas carbonate reservoir. 2024;**102**(9): 3103–18.
11. Shafiq MU, Mahmud HKB, Arif M. Mineralogy and pore topology analysis during matrix acidizing of tight sandstone and dolomite formations using chelating agents. *Journal of Petroleum Science and Engineering*. 2018;**167**: 869–76.
12. Elsheikh E, Moawad T, Alnetaifi A, AlQuraishi A, Almutawea Y. An evaluation of organic acids implementation in stimulating tight Saudi Reservoirs: A case study on Sarah sandstone formation. *Journal of King Saud University-Engineering Sciences*. 2022;**34**(7): 352–8.

13. Dargi M, Khamsehchi E, Kalatehno JM, Soury F, editors. Simulation-Based Optimization of Gas Storage Reservoir Through Acidizing. *SPE EOR Conference at Oil and Gas West Asia*; 2025: SPE.
14. Abdollahi R, Esfandyari H, Pari MN, Davarpanah A. Conventional diverting techniques and novel fibr-assisted self-diverting system in carbonate reservoir acidizing with successful case studies. *Petroleum Research*. 2021;**6**(3): 247–56.
15. Garrouch AA, Jennings AR Jr. A contemporary approach to carbonate matrix acidizing. *Journal of Petroleum Science and Engineering*. 2017;**158**: 129–43.
16. Adewunmi AA, Solling T, Sultan AS, Saikia T. Emulsified acid systems for oil well stimulation: A review. *Journal of Petroleum Science and Engineering*. 2022;**208**: 109569.
17. Chacon OG, Pournik M. Matrix acidizing in carbonate formations. *Processes*. 2022;**10**(1): 174.
18. Al-Shargabi M, Davoodi S, Wood DA, Ali M, Rukavishnikov VS, Minaev KM. A critical review of self-diverting acid treatments applied to carbonate oil and gas reservoirs. *Petroleum Science*. 2022.
19. Barri AA, Hassan AM, Aljawad MS, Mahmoud M. Effect of treatment conditions on matrix stimulation of carbonate rocks with chelating agents. *Arabian Journal for Science and Engineering*. 2021: 1–14.
20. Mahmoud M. Reaction of chelating agents and catalyst with sandstone minerals during matrix acid treatment. *Arabian Journal for Science and Engineering*. 2018;**43**: 5745–56.
21. Abbasi A, Khamsehchi E, Khaleghi MR, Mahdavi Kalatehno J, Panjalizadeh HJJoPE, Technology P. A new formulation for removing condensate blockage for low permeable gas reservoir. 2024;**14**(8): 2491–507.
22. Letichevskiy A, Nikitin A, Parfenov A, Makarenko V, Lavrov I, Rukan G, et al, editors. Foam acid treatment-The key to stimulation of carbonate reservoirs in depleted oil fields of the Samara region. *SPE Russian petroleum technology conference*; 2017: OnePetro.
23. Gao J, Feng P, Wang D, Shao S, Cui B, Wang G, editors. Large Liquid Volume Deep Penetration Acidizing in Iraq Missan Oilfield-3 Case Study. *International Petroleum Technology Conference*; 2019: IPTC.
24. Al-Rekabi MA, Aktebanee A, Al-Ghaffari AS, Saleem T, editors. Carbonate matrix acidizing efficiency from acidizing induced skin point of view: case study in Majnoon oilfield. *International Petroleum Technology Conference*; 2020: IPTC.
25. Alvarez J, Alarcon X, Tellez J, Sameer Y, Soran M, Rebeen Q, editors. Implementing a New Foam-Acid Technology for Matrix Stimulation of Challenging Low Pressure, Naturally Fractured Carbonates Reservoirs: Case Studies, Northern Iraq. *International Petroleum Technology Conference*; 2022: IPTC.
26. Shajari M, Malhani H. Stimulation of Low Pressure Carbonates Reservoirs with a Triphase Emulsified Acid System" 3PEA" of Southern Iran-Case Studies. *Improved Oil and Gas Recovery*. 2023; 7.
27. Keihani Kamal M, Mahdavi Kalatehno J, Daneshfar P, Yousefmarzi FJSR. A comprehensive analysis of carbonate matrix acidizing using viscoelastic diverting acid system in a gas field. 2024;**14**(1): 1499.
28. Mahdavi Kalatehno J, Khamsehchi E, Keihanikamal M, Yousefmarzi F, Dargi M, Daneshfar PJJJoPE, et al A successful case study of using HCl and viscoelastic diverting acid systems for carbonate matrix acidizing in an oil well with optimized predictive model. 2025;**15**(1): 1.
29. Alhamad L, Alrashed A, Al Munif E, Miskimins J, editors. A review of organic acids roles in acidizing operations for carbonate and sandstone formations. *SPE International Conference and Exhibition on Formation Damage Control*; 2020: OnePetro.

30. Shirazi MM, Ayatollahi S, Ghotbi C. Damage evaluation of acid-oil emulsion and asphaltic sludge formation caused by acidizing of asphaltenic oil reservoir. *Journal of Petroleum Science and Engineering*. 2019;**174**: 880–90.
31. Mirvakili A, Rahimpour M, Jahanmiri A. Experimental study of iron-control agents selection for high temperature sour gas and oil wells acidizing process. *Chemical Engineering Research and Design*. 2012;**90**(11): 1823–33.
32. Chen Y, Yang W. Formulation of corrosion inhibitors. *Water chemistry*. 2019: 1–22.
33. Kalatehno JM, Khamsehchi E. A novel packer fluid for completing HP/HT oil and gas wells. *Journal of Petroleum Science and Engineering*. 2021;**203**: 108538.
34. Ismail A, Irshad H, Zeino A, Toor I. Electrochemical corrosion performance of aromatic functionalized imidazole inhibitor under hydrodynamic conditions on API X65 carbon steel in 1 M HCl solution. *Arabian Journal for Science and Engineering*. 2019;**44**(6): 5877–88.
35. Hossain SZ, Al-Shater A, Kareem S, Salman A, Ali R, Ezuber H, et al Cinnamaldehyde as a green inhibitor in mitigating AISI 1015 carbon steel corrosion in HCl. *Arabian Journal for Science and Engineering*. 2019;**44**: 5489–99.
36. Amosa M, Mohammed I, Yaro S. Sulphide scavengers in oil and gas industry—a review. *Nafta*. 2010;**61**(2): 85–92.
37. Al-Duailej YK, Al-Mutairi SH, Al-Humaidan AY, editors. Evaluation of triazine-based H₂S scavengers for stimulation treatments. *SPE/DGS Saudi Arabia Section Technical Symposium and Exhibition*; 2010: OnePetro.